

VOIS Innovation Marathon 2025-2026

Project Title:
GlacierTide - Incremental Analysis of Glacier Melt
and its Impact using Machine Learning and
Interactive Visualization Tools

TEAM NO:91

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Project Details:

GlacierTide is an artificial intelligence–based climate intelligence platform that analyzes, predicts, and visualizes the impact of glacier melt on global and regional sea levels. The system leverages machine learning models—including Linear Regression, Decision Trees, Random Forest, and XGBoost—trained on Global Mean Sea Level (GMSL) data from 1993 to 2024. To reduce short-term prediction error, the platform integrates a real-time calibration pipeline using live NASA and NOAA satellite data, accounting for thermal expansion, ice mass contribution, vertical land movement, and sea surface temperature anomalies.

A key innovation of GlacierTide is the integration of Retrieval-Augmented Generation (RAG), where all predictive insights are scientifically grounded using semantically retrieved evidence from IPCC AR6 reports stored in a vector database. The platform is implemented as a full-stack web application with a MERN-based frontend, a Flask-based ML inference service, and interactive 2D and 3D visualizations that demonstrate region-specific sea-level rise impacts across six vulnerable coastal regions. GlacierTide bridges the gap between climate science research and policymaking by delivering predictive, explainable, and accessible climate intelligence.

Problem Statement:

The rates at which the global temperatures are increasing have been unprecedented and the total amount of CO₂ in the atmosphere has surged to more than 420 ppm in 2024. The effects of this warming cause a set of climate tipping points: Arctic sea ice melts, permafrost melts, and glaciers melt non-linearly. The ice cover in Greenland and West Antarctic becomes unstable and poses the risk of sudden, meter-scale sea-level rise as opposed to centimetre-high rise. Conventional climate platforms suffer from critical scientific gaps, including a reliance on static, archaic datasets and being computationally inaccessible to many. Existing “black-box” machine learning models lack explainability and trust, providing predictions without verifiable scientific reasons. This complex data is often trapped in dense technical reports, creating a “visualization deficit” that hinders stakeholder engagement and provides low levels of actionable decision support. This project, “GlacierTide,” addresses this research gap by creating an accessible, AI-driven platform. It integrates ensemble ML, real-time data fusion, and RAG-grounded scientific validation to provide predictions that are not only accurate but also transparent and understandable.

Need of Project:

- Global Crisis: Glaciers melting at unprecedented rates with severe sea level rise impacts on coastal populations.
- Accessibility Gap: Existing climate models are computational bottlenecks, excluding non-expert stakeholders and policymakers.
- Technology Gap: Lack of interactive, real-time, and explainable AI-driven climate insights for decision-making.
- Decision Support Gap: No accessible platform integrating ML ensemble predictions, satellite data, and peer-reviewed validation.
- GlacierTide Solution: Interactive AI platform with ensemble ML models, real-time calibration, and RAG-based scientific validation to enable informed climate action.

Proposed Solution:

The product is a proposal under the name GlacierTide, which is an all-encompassing AI-enabled climate intelligence platform aimed at providing accurate, explainable, and actionable information about sea-level rise caused by glaciers. The system is developed based on a four-layer system of ML Ensemble Layer, Real-Time Calibration Layer, RAG Scientific Validation Layer, and AI Agent Orchestration Layer. The ML Ensemble Layer is a combination of four predictive models, each with high predictive performance such as Linear Regression, Decision Tree, Random Forest, and XGBoost with a $R^2 = 0.92$ and a mean absolute error of 5 mm. The Real-Time Calibration Layer combines real-time NASA and NOAA API with a 30-minute refresh rate and uses regional multipliers via a dual multiplier system, resulting in a 20-30 percent better balance between short-term prediction accuracy and the expensive, difficult to implement, 2-hour refresh rate model. In order to achieve scientific transparency and trust, the RAG Scientific Validation Layer bases each prediction on semantically retrieved evidence in 1,557 IPCC AR6 document chunks stored in a ChromaDB vector database. Lastly, The AI Agent Orchestration Layer is based on a ReAct architecture and is powered by a Groq-hosted Llama 3.3 (70B) model. It provides a conversational interface with context memory, allowing users to interactively query, explore, and understand climate risks. These layers allow users to make informed climate decisions, which are explainable and scalable, and can be done through collaboration between these layers.

Technology Used:

Backend Technologies:

- Flask
- Python(Scikit-Learn, XGBoost, Pandas, NumPy)
- ChromaDB, MongoDB
- Express, Mongoose
- SendGrid

Frontend Technologies:

- ReactJS, Axios,
- Tailwind CSS, Vite,
- Three.js, Recharts.js, Leaflet.js

AIML Technologies:

- Machine Learning (Linear Regression, Decision Tree, Random Forest, XGBoost)
- LangChain (RAG pipeline, ReAct agent)
- HuggingFace
- GroqCloud
- NASA & NOAA APIs

Project Outcomes:

GlacierTide Architecture - Four Layers

- ◆ ML Ensemble Layer

Ensemble of 4 models (Linear, Decision Tree, Random Forest, XGBoost)

$R^2 = 0.92$ | MAE = 5mm

- ◆ Real-Time Calibration Layer

NASA/NOAA API integration | 30-min refresh | Regional multipliers

Dual multiplier system: +20-30% accuracy improvement

- ◆ RAG Scientific Validation Layer

1,557 IPCC AR6 document chunks | ChromaDB vector database

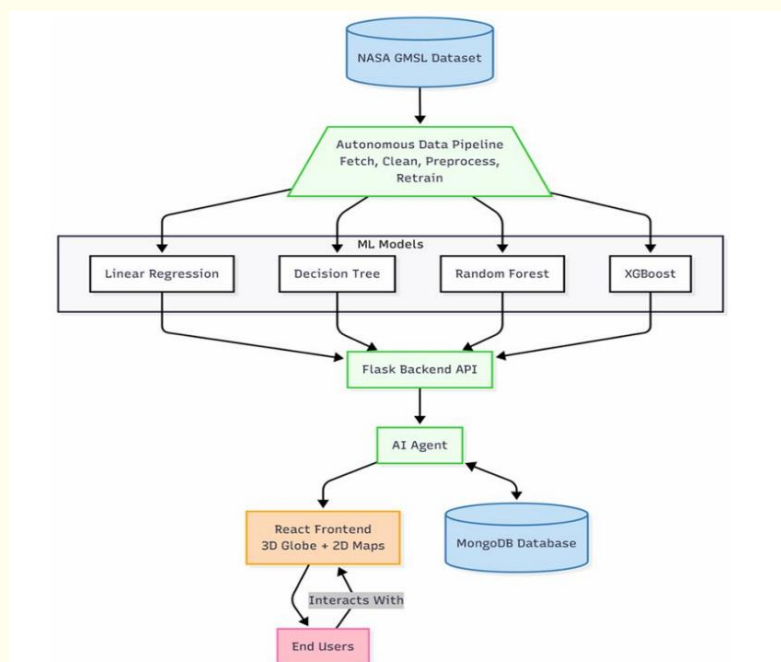
Every prediction grounded in peer-reviewed sources

- ◆ AI Agent Orchestration Layer

Groq Llama 3.3 70B | ReAct-based architecture

Conversational interface with context memory

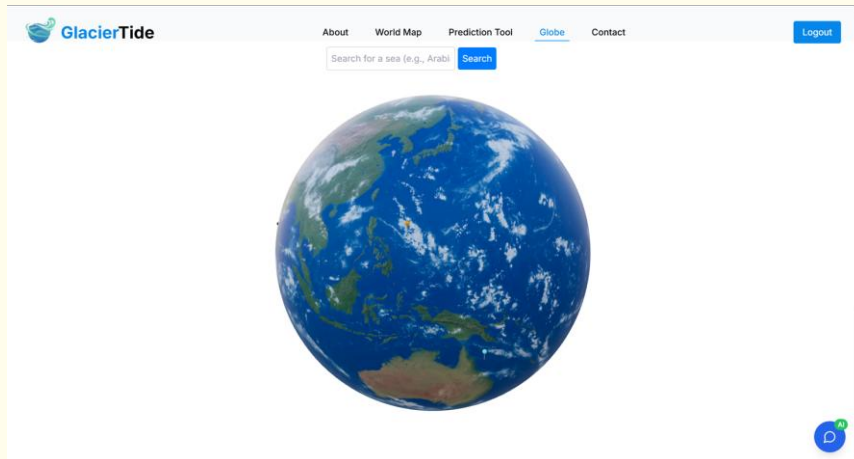
The steps can be visualized by watching the following system architecture:



Results:

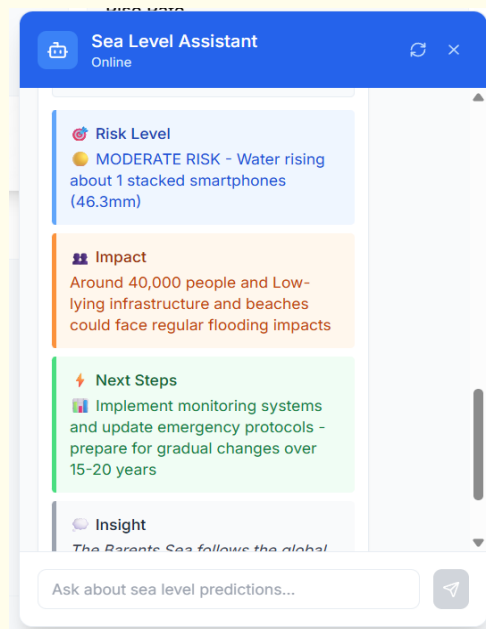
Graphical User Interface (GUI):

The GUI provides interactive dashboards and geospatial visualizations for region-specific sea-level rise analysis.



AI Agent:

The AI agent enables natural-language queries on climate risks, providing explainable predictions supported by IPCC-backed evidence.



Future scope for project enhancement:

The proposed future scopes for GlacierTide are as follows:

1. Geographic Expansion
2. Increased Forecasting Capabilities
3. Economic Impact Module
4. Fine-Tuned Domain LLM
5. Multi-Modal Integration